Symbolic Logic Framework for Situational Awareness in Mixed Autonomy

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Contents

List of Figures 4

List of Abbreviations 5

Nomenclature 6

1 Introduction 7
- 1.1 Background . . . 7
  - 1.1.1 Participatory Design of eHMIs . . . 8
  - 1.1.2 Crowd-Based Design . . . 8
  - 1.1.3 Communicating Awareness with eHMIs . . . 8
- 1.2 Summary . . . 8

2 Collaborative Crowdsourcing for external Human-Machine Interface Designs 9
- 2.1 Method . . . 9
  - 2.1.1 Expert Analysis and Feedback . . . 9
  - 2.1.2 Crowdsourced Design Collaboration . . . 9
- 2.2 Results . . . 10
  - 2.2.1 Iteration 1 (26 participants) . . . 11
  - 2.2.2 Iteration 2 (19 participants) . . . 12
  - 2.2.3 Iteration 3 (14 participants) . . . 12
  - 2.2.4 Iteration 4 (7 participants) . . . 12
- 2.3 Final Crowdsourced eHMI Designs . . . 13
  - 2.3.1 Popular-design . . . 13
  - 2.3.2 Innovative-design . . . 13
  - 2.3.3 Baseline Design . . . 14
  - 2.3.4 Feature Activation in Interface Designs . . . 14
- 2.4 Conclusion . . . 14

3 Evaluation of Crowdsourced eHMI Concepts 15
- 3.1 Experiment Design . . . 15
  - 3.1.1 Conditions . . . 15
  - 3.1.2 Measures . . . 15

3.1.3 Hypotheses 16
3.1.4 Simulation Environment 16
3.1.5 Procedure 16
3.2 Results 17
3.2.1 Response Time 17
3.2.2 User Experience 17
3.2.3 Qualitative eHMI interpretation 18
3.3 Conclusion 18

List of Figures

2.1 Illustration of the *Expert Analysis and Feedback* process. Participant-submitted sketches are grouped into similar clusters (*concepts*). An independent expert rates each concept's effectiveness and feasibility, providing justifications. These illustrated concepts, with descriptions and feedback, form the next stage of the visualization tree, guiding the next iteration. 10
2.2 Illustration of the *Crowdsourced Design Collaboration* process. Participants iteratively refine ideas inspired by earlier contributions. Using a tree-based visualization, top-level concepts branch into subsequent sketches. Participants review and rate prior concepts and then submit their contribution. Each submission includes a sketch and a brief explanation. 11
2.3 Concepts developed in the **first** collaborative iteration. All concepts have a brief descriptions and expert feedback, along with expert ratings for effectiveness (∘) and feasibility (♢), where 7 indicates the highest effectiveness and feasibility. 11
2.4 Concepts developed in the **second**, **third** and **fourth** iterations. All concepts have brief descriptions and expert feedback, along with expert ratings for effectiveness (∘) and feasibility (♢), where 7 indicates highest effectiveness and feasibility. 12
2.5 Screenshots of the three eHMI designs created in simulation captured over the three main points of AV-Pedestrian communication: (Point A) Pedestrian Awareness, (Point B) Yielding Intention, and (Point C) Crossing Facilitation. 13

3.1 Configuration of the traffic scenario used for evaluating eHMI designs. Key points where specific cues were activated are highlighted: (Point A) communication of pedestrian detection, (Point B) signaling yielding intention, and (Point C) additional crossing facilitation. Sample screenshots from the simulation of an autonomous bus without eHMI are shown at the bottom. 16
3.2 Results for response times and user experience. Significance levels indicated as ***= $p<0.001$ and *= $p<0.05$. (Note on the participant key press plot: a higher percentage of participants pressing the key earlier indicates higher interpretability. 17

List of Abbreviations

eHMI external Human-Machine Interface

AV Autonomous Vehicle

UEQ User Experience Questionnaire

Nomenclature

M= Mean
SD= Standard Deviation
z= z-value
p= p-value

1. Introduction

This report is the deliverable for Task 4.3 in the third year of the SymAware project. It is based on the following research contribution:

1. Ronald Cumbal, Marcus Göransson, Alexandros Rouchitsas, Didem Gürdür Broo, and Ginevra Castellano. 2026. A Collaborative Crowdsourcing Method for Designing External Interfaces for Autonomous Vehicles. In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI'26), April 13-17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 17 pages. https://doi.org/10.1145/3772318.3791228

This report corresponds to Work Package 4 (WP4), Task 4.3: "Gathering and Communicating Spatial Information to Humans." Building on the human-centred design approach taken in Task 3.2, we propose a scalable participatory method for designing external Human-Machine Interfaces (eHMIs) for Autonomous Vehicles (AVs). The method uses a collaborative, iterative crowdsourcing process in which participants submit sketched interface ideas, receive expert feedback, and build upon one another's contributions across multiple iterations.

This work addresses how learning-enabled agents can communicate complex spatial information in dynamic urban settings. Interfaces are designed to reflect the agent's **knowledge and awareness goals**, enabling the AV to translate its internal environmental mapping into intuitive visual cues. We evaluate how **spatial data**, such as navigation intent and hazard proximity, can be conveyed through eHMIs to improve shared situational awareness and human-agent interaction quality. Designs are assessed using established performance metrics and qualitative methods, with a focus on pedestrians as a critical user group for AV integration in urban environments.

1.1 Background

Participatory design methods are well-established for involving stakeholders in technology development [1, 2], yet remain resource-intensive and limited in scale. Crowdworking platforms offer a scalable alternative for distributed problem-solving [3, 4], and crowdsourcing has demonstrated value in both product design [5, 6, 7] and user-centered technology design [8]. However, conventional crowdsourcing tasks tend to favour pragmatic outputs over creative exploration. Our approach addresses this by embedding an iterative, collaborative structure that encourages speculative ideation, enabling participants to reflect on and shape how spatial information might be gathered and communicated by emerging autonomous systems.

1.1.1 Participatory Design of eHMIs

Large-scale surveys have provided useful insights into public expectations of AVs [9], while co-design sessions have enabled deeper engagement but at the cost of scale [10, 11, 12]. Noteworthy efforts include the CityMobil2 project [13] and studies foregrounding underrepresented groups such as people with mobility or visual impairments [14, 15, 16]. Relevant to the method proposed in this report, Mahadevan et al. [17] applied the PICTIVE method to generate 34 unique eHMI concepts, and Lakhdhir et al. [11] extended this to pedestrian-worn interfaces. These studies demonstrate the value of pedestrian involvement in ideation, yet their limited sample sizes restrict generalisability—a gap our scalable approach is designed to address.

1.1.2 Crowd-Based Design

Crowdsourcing harnesses distributed online communities for creative and analytical tasks [3, 4]. In design contexts, crowd feedback has been shown to improve creative output [18, 19], including at both superficial and conceptual levels when non-experts are involved [20]. Expert input remains essential for ensuring technical feasibility, and combining crowd and expert contributions has been shown to yield higher-quality results [21]. Although crowd-based design has rarely been applied to technical domains, structured guidance and expert feedback can bridge this gap: harnessing collective imagination while maintaining rigour.

1.1.3 Communicating Awareness with eHMIs

A core function of eHMIs is to support pedestrian decision-making by conveying a vehicle's position, detections, and intentions. Research consistently shows that pedestrians prioritise confirmation of detection [13, 22, 23], followed by intent cues such as yielding or proceeding [17, 11]. Communication strategies explored in the literature include 360-degrees displays [24], light strips [25, 26], icons [27], anthropomorphic cues [28, 29, 24, 30], text messages [31], and multimodal combinations [17].

Effective communication in these environments requires agents with knowledge awareness goals, enabling interfaces to more accurately translate the vehicle's spatial understanding into legible cues. These challenges motivate continued user-centred exploration of how spatial information about AV intent can be gathered from users and effectively communicated back to them.

1.2 Summary

This introduction addresses the challenge of **gathering and communicating spatial information** between AVs and pedestrians in urban environments. Participatory design methods are effective for involving users in this process but are typically limited in scale, while crowdsourcing platforms offer breadth but lack structured creative engagement. Our approach combines both by embedding an iterative, expert-guided crowdsourcing process that fosters collective ideation around eHMI design. Prior work in participatory eHMI design and crowd-based methods supports the feasibility of this combination, provided that expert feedback ensures technical quality. Research on AV-pedestrian communication further highlights that pedestrians require **clear spatial cues**, particularly confirmation of detection and vehicle intent, to navigate safely. The proposed method is positioned to scale user involvement in the design of such interfaces, generating diverse concepts that are grounded in user needs and evaluated against established performance criteria.

2. Collaborative Crowdsourcing for external Human-Machine Interface Designs

Our method integrates participatory design principles with scalable crowdsourcing techniques to engage users meaningfully in the design of spatial communication interfaces for autonomous vehicles. The approach is structured around two complementary components:

- Iterative Idea-building: Adapting Yu et al.'s framework [32], participants expand upon and combine prior submissions, fostering progressively original design outcomes. Expert feedback is incorporated at each cycle to guide contributions toward technically feasible solutions.
- Structured Visualization: Following Sun et al.'s tree visualization strategy [19], a tree-based structure organizes the evolution of ideas across iterations, with expert feedback bridging creative exploration and practical implementation.

2.1 Method

The process alternates between two stages: (1) crowdsourced design collaboration and (2) expert analysis with feedback that are presented next.

2.1.1 Expert Analysis and Feedback

Participant-generated sketches were clustered into conceptual groups, each standardized graphically and assessed by an independent expert in Human Factors of Automated Driving on two criteria: *effectiveness* (clarity of pedestrian awareness communication) and *feasibility* (practicality of implementation). These evaluations were recorded using a 7-point Likert scale, each accompanied by short justifications. Summarized versions of the feedback were then embedded into the visualization tree as parent nodes for subsequent iterations (as illustrated in Figure 2.1).

2.1.2 Crowdsourced Design Collaboration

Building on expert-reviewed concepts, participants engaged in collaborative cycles of refinement, combination, and reimagination. The website used to collect participant contributions was developed using jsPsych and hosted on Cognition. After providing consent and completing a demographic

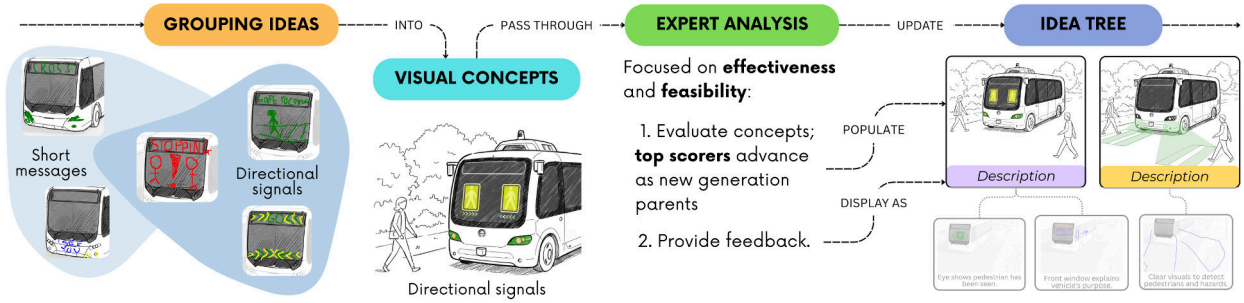

Figure 2.1: Illustration of the *Expert Analysis and Feedback* process. Participant-submitted sketches are grouped into similar clusters (*concepts*). An independent expert rates each concept's effectiveness and feasibility, providing justifications. These illustrated concepts, with descriptions and feedback, form the next stage of the visualization tree, guiding the next iteration.

questionnaire, participants were introduced to the study goals and the importance of AV-pedestrian spatial communication.

Two scenarios differing in perceived risk were used to prompt reflection on different dimensions of spatial information exchange:

- **Low-Risk Scenario**: A pedestrian crosses at a signalized zebra crossing as an AV approaches; the situation is predictable and the AV is expected to comply with traffic signals [13].
- **High-Risk Scenario**: A pedestrian crosses in a crowded area without clear signals; the AV is closer and its intentions are less certain.

Participants first engaged with the low-risk scenario and were asked how they would like the AV to communicate its spatial awareness. They then rated five randomly selected designs for creativity (7-point Likert scale). For the high-risk scenario, seven reflective prompts (e.g., "*What if two children step onto the street from opposite sides in heavy rain?*") encouraged consideration of real-world complexity. Participants then created their own design using an embedded drawing interface, explained their rationale, and indicated whether their idea built on existing concepts or was entirely new. A summarized version of this process is shown in Figure 2.2.

A *stopping criterion* was applied: each submission was assigned a novelty score (5 = new modality, 3 = substantial refinement, 1 = minor tweak), and data collection ceased when the average novelty score of the last five participants fell below 2 and no new modality appeared in three consecutive submissions.

2.2 Results

Participants were recruited via `Prolific` (European residents, English proficient). A total of 67 participants submitted designs sketches throughout four process iterations. The participants had an average age of 37.86 (SD: 11.82), with gender distribution of 1 non-binary, 31 male, and 35 female participants. Out of these, 42 used a private vehicle for transportation, 14 Public transportation and 11 Walking/Cycling. Participants were from 13 European countries, with the United Kingdom being the largest sample (38), then Poland (7), Italy (4), and Germany (4) accounting for the

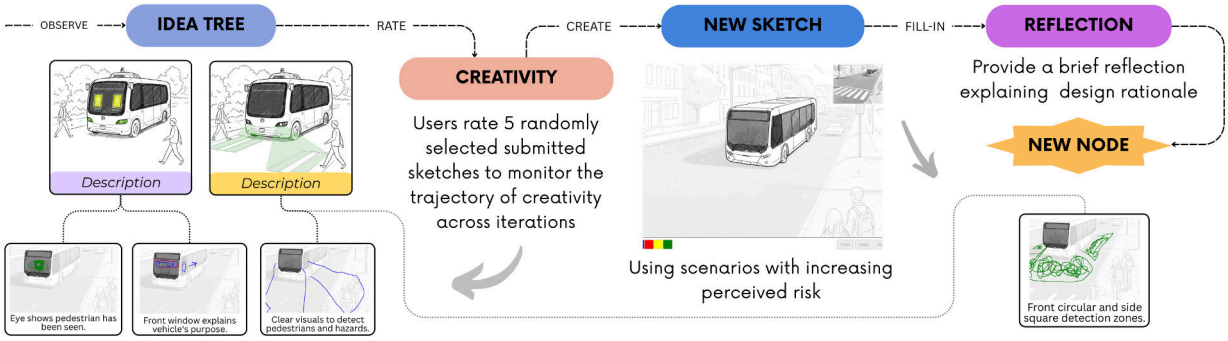

Figure 2.2: Illustration of the *Crowdsourced Design Collaboration* process. Participants iteratively refine ideas inspired by earlier contributions. Using a tree-based visualization, top-level concepts branch into subsequent sketches. Participants review and rate prior concepts and then submit their contribution. Each submission includes a sketch and a brief explanation.

majority. The median study duration was 17 minutes, and participants were compensated at £12.91 per hour.

Thematic Analysis Submissions were analyzed using reflexive thematic analysis [33]. Open coding of reflective and descriptive responses yielded 18 design codes organized into five themes: *lights*, *displays/screens*, *road projection*, *audio*, and *novel ideas*. Seven motivational themes were identified: *adaptiveness*, *awareness*, *familiar conventions*, *accessibility*, *intent transparency*, *safety*, and *visibility*. **Note:** In the following results, we report the number of participants who mentioned each theme in each interaction round, counting multiple mentions by the same participant once.

2.2.1 Iteration 1 (26 participants)

Light-based signals were the most common proposal (18/26), including: **Concept B1** (omnidirectional 360-degrees LED bands); **Concept B2** (directional spotlights tracking pedestrians); and **Concept B3** (live pedestrian images on the AV's external screen as bi-directional acknowledgment). Road and pavement projections appeared in 10/26 submissions (**Concept C1**: projected icons/text; **Concept C2**: directional lighting on nearby infrastructure). Audio and tactile signals

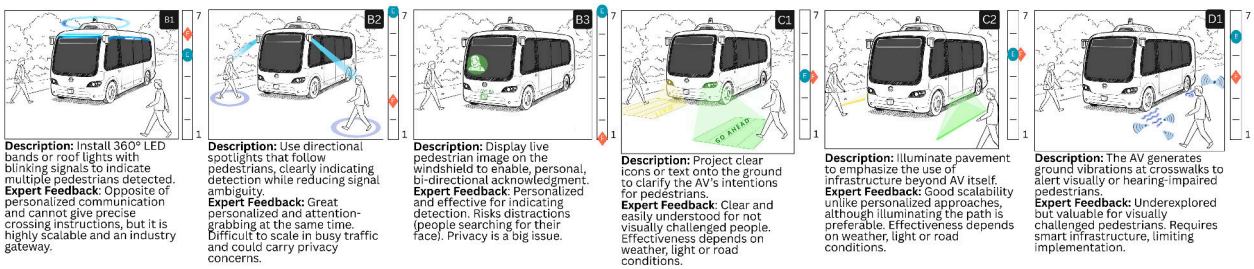

Figure 2.3: Concepts developed in the **first** collaborative iteration. All concepts have a brief descriptions and expert feedback, along with expert ratings for effectiveness (○) and feasibility (◇), where 7 indicates the highest effectiveness and feasibility.

were proposed by 11/26 participants, with **Concept D1** introducing ground vibrations at crosswalks for inclusive spatial warning. Front displays communicating vehicle intent were mentioned by 12/26 participants. The illustration of these concepts are shown in Figure 2.3.

2.2.2 Iteration 2 (19 participants)

Lights remained dominant (15/19). Novel concepts included dynamic safe/unsafe zone demarcation using dual-color lights and future path prediction (**Concept C3**). New modalities introduced: **Concept E** (a robot bus driver exiting the vehicle to reassure pedestrians); **Concept F** (Bluetooth messaging for detection and warning); and **Concept G** (physical barriers or flags auto-deploying to prevent unsafe crossings). These are illustrated in Figure 2.4

2.2.3 Iteration 3 (14 participants)

This iteration was completed with 15 participants; however, the data from one participant was excluded due to responses that reflected an extremely negative—and at times offensive—stance toward the implementation of autonomous vehicles.

Light-based signals remained dominant (13/14). Audio cues featured prominently (6/14), with emphasis on voice clarity. One participant proposed displaying the autonomous vehicle's predicted future path (**Concept C4**). **Concept H** introduced eye-like animations to convey the autonomous vehicle's spatial awareness of pedestrians.

2.2.4 Iteration 4 (7 participants)

The final iteration, after applying stopping criteria, involved seven participants and reinforced earlier trends. This iteration reinforced established trends: lights (6/7) and audio/voice (4/7)

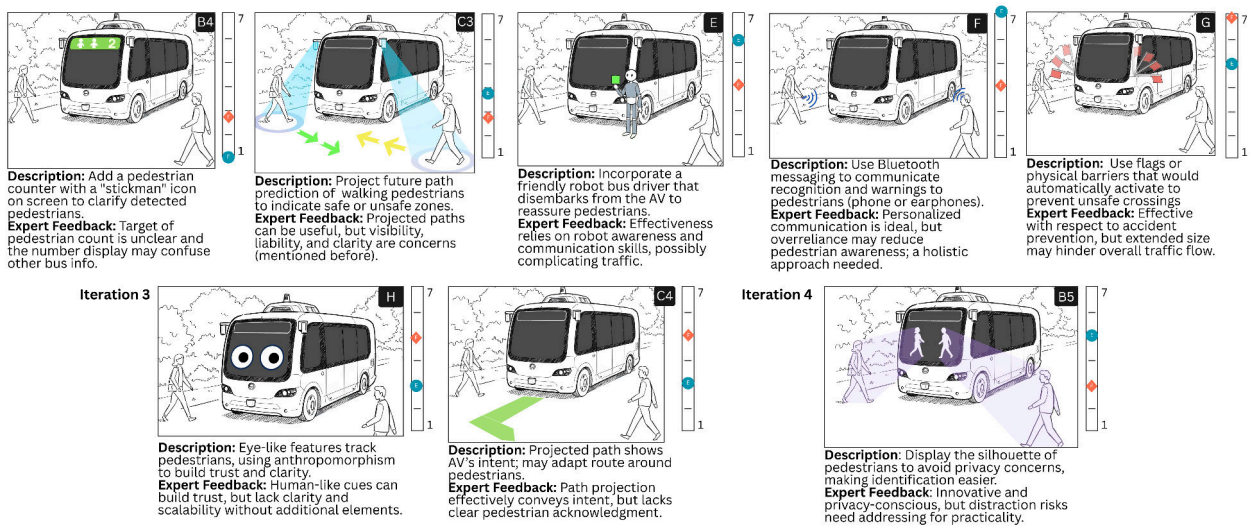

Figure 2.4: Concepts developed in the **second**, **third** and **fourth** iterations. All concepts have brief descriptions and expert feedback, along with expert ratings for effectiveness (○) and feasibility (◇), where 7 indicates highest effectiveness and feasibility.

remained primary. **Concept B5** ("wall of light") emerged as a privacy-preserving visualization projecting pedestrian silhouettes to confirm detection without identifying individuals.

2.3 Final Crowdsourced eHMI Designs

Two final eHMI designs were derived from the collaborative process, complemented by a state-of-the-art baseline for evaluation. The designs are shown in Figure 2.5.

2.3.1 Popular-design

This design reflects the most frequently proposed spatial communication strategies: a *360-degree color-coded light* band (steady green = safe to cross), *ground projection* of a zebra crosswalk to indicate a safe path, and an *icon display* to signal pedestrian detection. *Auditory cues*, while frequently proposed, were excluded due to the video-based simulation environment.

2.3.2 Innovative-design

This design was identified by averaging expert effectiveness and feasibility ratings across concepts and selecting the highest-scoring novel ideas. The design combines a green *360-degree LED band* to indicate yielding with a *future path projection* to convey pedestrian awareness, complemented by *directional spotlights* (Concept B2), also rated highly , to further strengthen communication.

Figure 2.5: Screenshots of the three eHMI designs created in simulation captured over the three main points of AV-Pedestrian communication: (Point A) Pedestrian Awareness, (Point B) Yielding Intention, and (Point C) Crossing Facilitation.

2.3.3 Baseline Design

This design uses a 360-degree display with eye-inspired circles that rotate toward detected pedestrians and blink to signal acknowledgment [24], paired with a cyan light band pulsing at 0.8 Hz to communicate yielding [34].

2.3.4 Feature Activation in Interface Designs

Features are triggered at specific spatial thresholds (see Figure 3.1 and Figure 2.5):

- **Point A (16 m) - Detection Acknowledgment** `popular-design` shows a pedestrian icon; `innovative-design` projects green zone rotating toward the pedestrian; and `baseline` activates "eyes," rotating toward pedestrian.
- **Point B (5 m) - Yielding Intention:** the light bands of the `popular-design` change from cyan to green; the light band of the `innovative-design` turns green; and the `baseline` begins pulsing its middle light band.
- **Point C - Crossing Facilitation:** the `popular-design` projects a zebra crossing; the `innovative-design` projects an arrow predicting walking direction; and the `baseline` remains unchanged from previous state.

2.4 Conclusion

This chapter presented a scalable, human-centred crowdsourcing method for the co-design of spatial communication interfaces for autonomous vehicles. The results demonstrate its effectiveness in advancing three core objectives of the project.

First, the method produced a rich set of interface concepts that support *mutual understanding between humans and autonomous agents*. Participants designed mechanisms through which AVs can communicate their spatial state and intentions –detection, yielding, and safe path guidance– while also proposing bi-directional features such as pedestrian count displays and acknowledgment signals, enabling the AV to demonstrate awareness of individual users.

Second, the iterative integration of expert feedback shaped contributions toward *transparent and explainable AV behaviour*. Participants progressively moved away from highly personalised or complex signals toward intuitive, convention-based spatial cues –such as standard traffic colors, symbolic projections, and privacy-preserving visualizations– that are interpretable without prior knowledge of AV systems. This convergence reflects a user-driven demand for legible, trustworthy communication.

Third, the collaborative structure kept *humans in the loop throughout the design lifecycle*. By enabling participants to build on one another's contributions across four iterations –57% of participants explicitly referenced prior submissions– the process functioned as a mechanism for collective sensemaking, allowing social norms and expectations to shape interface evolution. Expert feedback, rather than constraining this process, redirected creative exploration toward feasible and scalable implementations, supporting the system's long-term adaptability to social input.

Taken together, these outcomes validate collaborative crowdsourcing as a practical approach for generating human-centred spatial communication interfaces at scale, and provide a concrete empirical basis for the evaluation presented in the following chapter.

3. Evaluation of Crowdsourced eHMI Concepts

This chapter evaluates the spatial communication interfaces generated through the collaborative crowdsourcing process using established behavioral and experiential measures. Participants viewed video stimuli of simulated AVs featuring the three eHMI conditions, enabling controlled assessment of how effectively each design conveys spatial information —specifically pedestrian detection and yielding intent. While video-based studies lack full interactive immersion [35], they remain a widely used, safe, and cost-effective methodology [36, 37]. Evaluation focused on visual components, as controlling for audio modalities in online settings introduces significant confounds; future work should extend this to multimodal conditions.

3.1 Experiment Design

3.1.1 Conditions

- **Baseline:** The baseline combines a 360-degree display with eye-inspired circles rotating toward detected pedestrians [24] —recognized as scalable and visible across multiple studies [38, 39, 40, 41, 42, 43]— with a cyan light band pulsing at 0.8 Hz to signal yielding [34]. This pairing represents current best practices: the light band is the most prevalent eHMI concept in the literature [44, 45, 46, 47, 48, 49, 50, 51, 52], while the directional eyes convey spatial awareness.
- **Popular Design:** This design reflects the most frequently proposed user concepts: a 360-degree green light band, a pedestrian icon for detection acknowledgment, and a ground projection of a zebra crosswalk to indicate a safe crossing path.
- **Innovative Design:** This design integrates the highest-rated expert-evaluated user concepts: a green 360-degree LED band, a future path projection for directional spatial guidance, and directional spotlights tracking individual pedestrians.

3.1.2 Measures

Two criteria were applied. **Interpretability** was assessed via response time —the moment participants pressed a key indicating they felt safe to cross— a widely used proxy for comprehension speed in eHMI studies [46, 53, 54]. **User experience** was assessed using the short-form User Experience Questionnaire (UEQ-S) [55], administered after each stimulus.

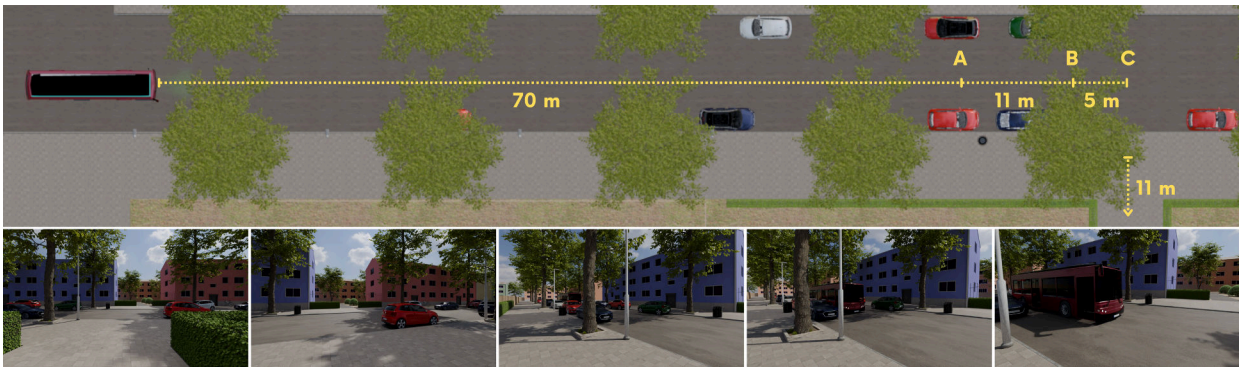

Figure 3.1: Configuration of the traffic scenario used for evaluating eHMI designs. Key points where specific cues were activated are highlighted: (Point A) communication of pedestrian detection, (Point B) signaling yielding intention, and (Point C) additional crossing facilitation. Sample screenshots from the simulation of an autonomous bus without eHMI are shown at the bottom.

3.1.3 Hypotheses

We hypothesized that (**H1**) crowdsourced designs would outperform the baseline in both interpretability (**H1.a**) and user experience (**H1.b**), given their grounding in intuitive, user-expressed expectations. We further hypothesized that (**H2**) the innovative-design would achieve the highest interpretability scores, due to the integration of expert-validated spatial communication strategies.

3.1.4 Simulation Environment

Videos were created in Blender depicting a high-risk urban scenario: no traffic signals, occluded visibility, and a crowded setting. Three 15-second experimental videos (one per condition) plus one practice video (no eHMI) were produced. In each, a bus begins moving about 85 m from the pedestrian, travels at 36 km/h, and decelerates at 20 m until stopping. The camera maintained a first-person pedestrian perspective at 1.27 m/s, with a brief left-right scan before reaching the curb. Figure 3.1 shows a top view of this configuration.

3.1.5 Procedure

The experiment was implemented in jsPsych and hosted on Cognition, restricted to devices with at least 280x720 pixel resolution. After consent and demographics, participants completed two practice trials before the main session. For the main task, participants were instructed to press the `Enter` key at the moment they felt it was safe to cross the street while watching each video stimulus. To familiarize them with the procedure, two practice trials were presented. First, a low-risk scenario with a different vehicle, to make sure participants were pressing the correct key. Videos were presented in randomized order; after each, participants completed the UEQ-S and provided a brief text interpretation of the design. No prior explanation of the eHMIs was given, ensuring spontaneous, ecologically valid responses [56, 57].

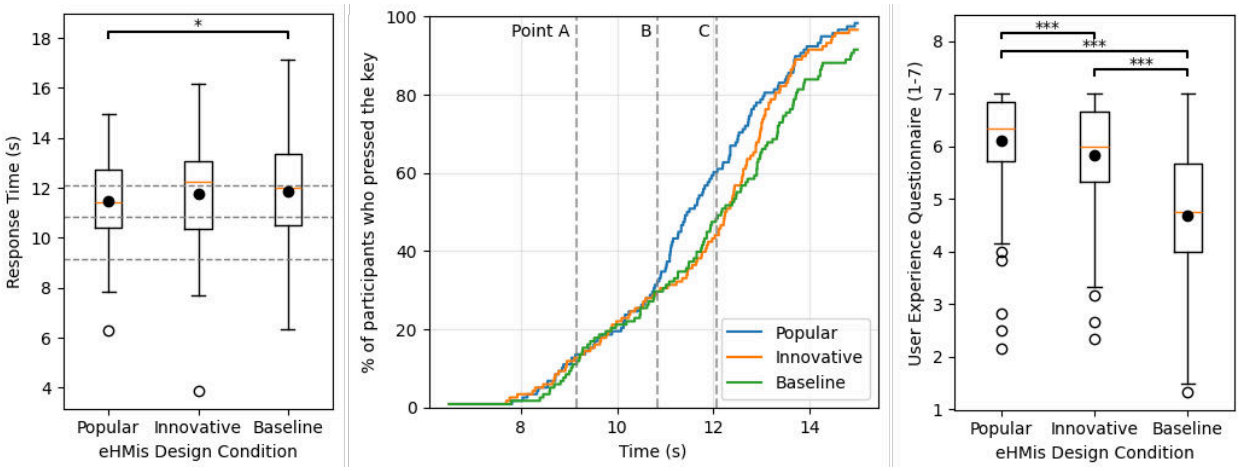

Figure 3.2: Results for response times and user experience. Significance levels indicated as ***= p<0.001 and *= p<0.05. (Note on the participant key press plot: a higher percentage of participants pressing the key earlier indicates higher interpretability.

3.2 Results

Of 131 recruited participants (European residents), 13 were excluded for two or more missing key presses. The final sample had a mean age of 38.2 years (SD: 12.81; 59 female, 58 male, 1 non-binary). AV acceptance was slightly above neutral (M: 3.15, SD: 0.98). Median study duration was 10 minutes; compensation was £8.67 per hour.

3.2.1 Response Time

A repeated-measures ANOVA revealed a significant main effect of eHMI condition, $F(2, 216) = 3.80, p = .024, \eta^2 = .009$, with Greenhouse-Geisser corrected degrees of freedom ($\varepsilon = 0.975$). Post-hoc Bonferroni-corrected t-tests showed that *response times were significantly faster for the* `popular-design` than for the `baseline` condition, $t(108) = -2.77$, $p = .020$, $Hedges'g = -0.23$, $BF_{10} = 3.93$. No significant differences were found between the `popular-design` and `innovative-design`, or between the `innovative-design` and `baseline`. These results are illustrates in left side of Figure 3.2.

The center of Figure 3.2 also shows response times across the full video sequence by plotting the percentage of participants who pressed the key. At point B (yielding intention), the `popular-design` shows a clear inflection, with more participants feeling safe, linked to the projection of green "safe zone", a feature frequently proposed during the crowdsourcing phase. At point C (crossing facilitation), the `innovative-design` produces a sharp increase, reflecting the use of an arrow projection of the pedestrian's path, which also echoes recurring symbol-based proposals from the ideation phase.

3.2.2 User Experience

A Friedman test revealed a significant effect of condition, $\chi^2(2) = 103.39, p < .001$. Post-hoc pairwise comparisons using Wilcoxon signed-rank tests with Bonferroni correction indicated a

significant difference between the `popular-design` and `innovative-design`, $W = 1512.0, p = .00037, Hedges'g = 0.26$; between `popular-design` and `baseline`, $W = 296.5$, $p < .001$, $Hedges'g = 1.23$; and between `innovative-design` and `baseline`, $W = 444.0$, $p < .001$, $Hedges'g = 0.97$. These results are illustrated in the right side of Figure 3.2.

3.2.3 Qualitative eHMI interpretation

The popular-design was consistently described as clear and intuitive —participants found the pedestrian icon, projected crosswalk, and green light band effectively communicated when it was safe to cross, reinforcing their sense of being acknowledged by the vehicle. The innovative-design was well received for its ground projections and directional arrows, though some participants noted brief initial uncertainty before the arrow appeared. The baseline performed worst: participants frequently reported relying on vehicle deceleration rather than the eHMI signals, reflecting low confidence in the spatial information conveyed.

3.3 Conclusion

This evaluation assessed how effectively human-centred, crowdsourced spatial communication interfaces support pedestrian decision-making in AV interactions, examining both the speed of spatial information comprehension and the quality of the resulting interaction experience.

Across both measures, the crowdsourced designs consistently outperformed the state-of-the-art baseline, supporting H1. The popular-design achieved the fastest response times (supporting H1.a) and the highest user experience scores (supporting H1.b), while the innovative-design ranked second on both measures, providing no support for H2. The baseline, despite representing established practice, performed worst —suggesting that abstract visual elements alone are insufficient for communicating spatial information reliably.

The key differentiator was the use of familiar, symbol-based spatial cues. The pedestrian icon in the popular-design provided immediate, unambiguous confirmation of detection; in contrast, the innovative-design's tracking light alone was insufficient until the directional arrow appeared. This pattern confirms that interpretability of spatial information is significantly enhanced by standardized, recognizable signals rather than novel or abstract ones —a finding that aligns with participants' own design priorities during the ideation phase.

These results carry direct implications for the design of human-agent interaction systems for spatial communication. First, interfaces that convey spatial state using familiar conventions measurably reduce response latency and cognitive load, improving the speed and confidence with which humans interpret AV intent. Second, user experience —reflecting perceived clarity, comfort, and sense of acknowledgment— is a meaningful and sensitive indicator of interaction quality, and crowdsourced designs anchored in user expectations consistently outperform expert-only solutions on this dimension. Third, the convergence between what participants designed and what performed best in evaluation validates the use of participatory methods not merely for ideation, but as a reliable mechanism for producing effective spatial communication solutions. Minor refinements to the innovative-design —such as introducing pedestrian icons earlier in the interaction sequence— could close the remaining performance gap, offering a practical path toward combining human-centred creativity with expert-guided feasibility.

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